RESEARCH STUDY - FEBRUARY 2026

AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors:
US/Europe Divergence Trajectories 2024-2030

Integrated Geostrategic and Economic Analysis - ECONOMETRIC ANNEX

76.9% of global operational AI compute = USA	**1.59x energy cost EU/US (PPP-adjusted)**	**3.46:1 CACI US/EU ratio (Power Mode)**

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ECONOMETRIC ANNEX

Empirical Validation of CACI through Panel Data

This annex presents the econometric validation of the CACI (Compute-Adjusted Competitiveness Index) proposed in Chapter II. The objective is to test whether the CACI, constructed according to the geometric Power Mode formula F^0.40 x L^0.20 x R^0.15 / E^0.25, effectively predicts AI productivity differentials between countries. We use a panel of 12 countries over the 2020-2024 period (N = 60 observations) and estimate three specifications: OLS pooled, fixed effects (within estimator), and random effects (GLS).

A.1 Data Panel Construction

The panel covers 12 economies representing over 90 percent of global AI compute: USA, China, UK, Germany, France, Japan, South Korea, India, Canada, Netherlands, Brazil, and Sweden. The period (2020-2024) captures the AI deployment acceleration phase and the first export control measures.

CACI is calculated according to the consolidated Power Mode formula: CACI(r,t) = F(r,t)^0.40 x L(r,t)^0.20 x R(r,t)^0.15 / E(r,t)^0.25. Weights are F = 0.40 (compute), L = 0.20 (human capital), R = 0.15 (regulatory access), E = 0.25 (energy cost).

A.3 Main Results

The CACI coefficient is positive and statistically significant at the 1 percent level across all three specifications. The elasticity estimated by the fixed effects model (preferred) is 0.251: a 10 percent increase in CACI is associated with a 2.5 percent increase in sectoral AI productivity.

The within R-squared of 0.692 indicates that the CACI Power Mode explains nearly 70 percent of the intra-country variance in AI productivity.

A.9 Phys/Sov Extension: Jurisdictional Validation

The Phys/Sov extension decomposes factor F into two multiplicative components: F(r) = F_phys(r) x F_sov(r), where F_phys is physically installed compute and F_sov is the fraction under non-US jurisdiction.

While the US and China are fully sovereign (F_sov=100%), the EU is largely sovereign on cluster ownership (99.2%) but highly dependent on the cloud workload layer (only ~28% sovereign).

Table A.1. Panel variables and sources.

Source: Author's compilation.

Variable	Definition	Unit	Source
F(r,t)	Accessible installed AI FLOPs	PetaFLOPs (H100-eq)	Epoch AI, Hawkins et al. (2025)
E(r,t)	Data center energy cost (PPP-adj)	USD/MWh	Eurostat, EIA, IEA (2025)
PROD(r,t)	AI-intensive sector productivity gain	Annual % gain	McKinsey, IMF, Fed Board

Table A.2. Panel regression results.

*Robust standard errors (clustered by country) in parentheses. *** p < 0.01.*

Variable	M1: OLS	M2: FE	M3: RE
ln(CACI)	0.173*** (0.038)	0.251*** (0.075)	0.504*** (0.020)
N	60	60	60
R2 within	0.227	0.692	0.920

Notes

1. Hawkins et al. (2025), 'AI Compute Sovereignty', Oxford Internet Institute.
2. CACI Power Mode formula: F^0.40 x L^0.20 x R^0.15 / E^0.25.
3. Hausman test: chi2 = 13.91 (p = 0.001) -> Fixed Effects preferred.

Public dashboard: https://mo0ogly.github.io/America-First-IA/dashboard/

Repository: https://github.com/mo0ogly/America-First-IA